

The Hierarchy Is Geometric

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Why the Electron Is Light and Gravity Is Weak

1. The Problem

The Standard Model has a hierarchy problem. The Higgs mass is roughly 10^{17} times smaller than the Planck mass. If you compute quantum corrections to the Higgs mass, they diverge quadratically — every virtual particle in the vacuum pushes the mass toward the Planck scale. The observed value (125 GeV, not 10^{19} GeV) requires cancellations to one part in 10^{34} .

This apparent fine-tuning has driven forty years of theoretical physics. Supersymmetry (SUSY) proposes a partner for every particle, doubling the spectrum to cancel the divergences. Large extra dimensions propose that gravity is diluted by propagating in hidden spatial directions. The multiverse proposes that our universe is one of 10^{500} vacua, and we happen to live in one where the cancellation occurred. Each proposal introduces new structure — new particles, new dimensions, new universes — to explain why a single number is small.

BST derives the number. No new structure required.

2. The Derivation

The electron lives on the Shilov boundary $\Sigma = S^4 \times S^1/Z_2$ of the bounded symmetric domain $D_{IV}^5 = SO_0(5,2)/[SO(5) \times SO(2)]$. Gravity is a bulk phenomenon — it operates throughout the interior of the domain. To couple a boundary excitation (the electron) to the bulk (gravity), the electron must traverse $C_2 = 6$ embedding layers:

$$\text{Boundary} \rightarrow D_{IV}^1 \rightarrow D_{IV}^2 \rightarrow D_{IV}^3 \rightarrow D_{IV}^4 \rightarrow D_{IV}^5 \rightarrow \text{gravity}$$

Each layer costs α^2 — one round trip through the Bergman kernel (boundary $\rightarrow$ bulk $\rightarrow$ boundary), where each leg picks up a factor of $\sqrt{\alpha}$ from the kernel's radial decay. Six layers, α^2 per layer:

$$\frac{m_e}{m_{Pl}} = 6\pi^5 \cdot \alpha^{2C_2} = 6\pi^5 \cdot \alpha^{12}$$

Predicted: $m_e/m_{Pl} = 4.187 \times 10^{-23}$. Observed: 4.185×10^{-23} . Error: 0.032%.

The “large number” is $\alpha^{12} = (1/137.036)^{12} \approx 2.3 \times 10^{-26}$. It is not tuned. The 137 comes from the Wyler formula — a volume ratio on D_{IV}^5 that yields $\alpha = 1/137.036$ to five significant figures. The $12 = 2C_2$ comes from the Casimir eigenvalue $C_2 = n_C + 1 = 6$, a topological invariant of the domain. Both numbers are geometric. Neither is adjustable.

3. Gravity

The same mechanism explains why gravity is weak. Newton's gravitational constant:

$$G = \frac{\hbar c \cdot (6\pi^5)^2 \cdot \alpha^{24}}{m_e^2}$$

Predicted: $6.679 \times 10^{-11} \text{ m}^3/(\text{kg}\cdot\text{s}^2)$. Observed: 6.6743×10^{-11} . Error: 0.07%.

The exponent $24 = 2 \times 2C_2 = 4 \times 6$. Gravity couples TWO masses, each traversing all $C_2 = 6$ Casimir channels. Two bodies $\times$ six channels $\times \alpha^2$ per channel $= \alpha^{24}$.

Gravity is weak because it is a 24th-order process in α . Electromagnetism is a 1st-order process (one factor of α). The ratio:

$$\frac{Gm_p^2}{\hbar c} \sim \alpha^{24} \cdot (6\pi^5)^2 \sim 5.9 \times 10^{-39}$$

This is the famous “large number” — the ratio of gravitational to electromagnetic coupling for protons. Dirac noticed it in 1937. He spent his career searching for an explanation. The explanation is six embedding layers, squared for two-body coupling.

4. What This Replaces

Theorem. The hierarchy $m_e/m_{Pl} = \alpha^{C_2}$ follows from $C_2 = 6$ embedding layers on D_{IV}^5 , each costing α^2 . No free parameters.

Corollary. The following proposals are unnecessary:

1. Supersymmetry. SUSY doubles the particle spectrum to cancel quadratic divergences. BST has no quadratic divergences — the bounded symmetric domain provides a natural UV completion. The Bergman kernel is finite everywhere. No partners needed.
 2. Large extra dimensions. Models with extra spatial dimensions (ADD, Randall-Sundrum) propose that gravity appears weak because it propagates in hidden dimensions. BST derives the weakness from α^{24} on a KNOWN domain with KNOWN dimension (complex dimension 5, real dimension 10). The dimensions are not hidden — they are the spectral directions of D_{IV}^5 .
 3. Anthropic multiverse. The string landscape proposes $\sim 10^{500}$ vacua with random cosmological constants, and we inhabit one where the hierarchy is small enough for chemistry. BST derives the hierarchy from a UNIQUE domain — D_{IV}^5 is the only bounded symmetric domain of type IV with complex dimension 5 that satisfies all 21 uniqueness conditions (see the companion Working Paper Section 37.5). There is one vacuum. The hierarchy is what it is because the geometry is what it is.
 4. Technicolor / composite Higgs. Models where the Higgs is a bound state of new strong dynamics. BST derives the Higgs mass as $v\sqrt{(2/\sqrt{60})} = 125.11 \text{ GeV}$, where $60 = n_C!/2 = 5!/2$ and $v = m_p^2/(g\cdot m_e)$. The Higgs is the radial mode of D_{IV}^5 — its mass follows from the domain's quartic structure. No new strong sector needed.
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5. Predictions

If this derivation is correct:

1. No superpartners. The LHC has found no SUSY particles at any mass. BST predicts it never will. There are no superpartners because there is no supersymmetry.
2. No extra spatial dimensions. Tabletop gravity experiments and collider searches for Kaluza-Klein excitations should continue to find nothing.
3. The fine structure constant does not vary. $\alpha = 1/137.036$ is a volume ratio, not a running coupling evaluated at a particular energy. Any confirmed variation of α in quasar absorption spectra falsifies BST.
4. The Higgs quartic coupling is $\lambda = 1/\sqrt{60}$. Precision measurements at future colliders (FCC-ee, ILC) can test this.
5. The proton is absolutely stable. BST has no grand unification, no X bosons, no proton decay. $\tau_p = \infty$. Hyper-Kamiokande will test this over the next decade.

Each prediction is falsifiable. Each contradicts at least one major beyond-Standard-Model program. The data so far — no SUSY at LHC, no proton decay at Super-Kamiokande, no α variation in quasar spectra — is consistent with BST and inconsistent with the programs BST replaces.

6. What It Means

The hierarchy problem was never about fine-tuning. It was about coordinates.

In the Standard Model's coordinates — quantum field theory on flat Minkowski space — the Higgs mass receives divergent corrections from every energy scale up to the Planck mass. The corrections must cancel to absurd precision. This looks like fine-tuning because the coordinates make it look like fine-tuning.

In BST's coordinates — the Bergman kernel on D_{IV}^5 — the electron mass is a boundary excitation separated from the gravitational bulk by six geometric layers. Each layer costs α^2 . The total suppression is $\alpha^{12} \approx 10^{-26}$. There is nothing to cancel, nothing to tune, nothing to explain except the geometry itself. And the geometry is fixed by topology.

The hierarchy is not a problem. It is a measurement — of the depth of a bounded symmetric domain.

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AC classification: (C=1, D=0). The hierarchy is one geometric evaluation — read off the Casimir eigenvalue and count the layers. No iteration, no sequential dependence. The hardest problem in beyond-Standard-Model physics is depth 0.